<table>
<tr><td>Course Code:
CSE219</td><td colspan="2">Course Title: Big Data Analytics

Type of Course: Laboratory Integrated</td><td>L- T-P-C</td><td>1</td><td>0</td><td>4</td><td>3</td></tr>
<tr><td>Version No.</td><td colspan="7">2.0</td></tr>
<tr><td>Course Pre-requisites</td><td colspan="7">DDL, DML of SQL Queries and Creation of Class & object, interface, reading & writing a file, control statements in java programming.</td></tr>
<tr><td>Anti-requisites</td><td colspan="7">NIL</td></tr>
<tr><td>Course Description</td><td colspan="7">This course is designed to provide the fundamental knowledge to equip students being able to handle real world big data problems including the three key resources of Big Data: people, organizations, and sensor. With the advancement of IT storage, processing, computation and sensing technologies, big data has become a novel norm of life.</td></tr>
<tr><td>Course Objective</td><td colspan="7">The objective of the course is to familiarize the learners with the concepts of Big Data Analytics and attain SKILL DEVELOPMENT through EXPERIENTIAL LEARNING techniques</td></tr>
<tr><td>Course Out Comes</td><td colspan="7">On successful completion of the course the students shall be able to:
1: Describe the fundamental concepts of big data analytics (Knowledge)
2: Apply Map-Reduce programming on the given datasets to extract required insights. (Application).
3: Employ appropriate Hadoop Ecosystem tools such as Hive, Hbase to perform data analytics for a given problem (Application)
4: Use Spark and nosql tool to analyse the given dataset for a given problem. (Application).</td></tr>
<tr><td>Course Content:</td><td colspan="7"></td></tr>
<tr><td>Module 1</td><td>Introduction to Big data Analytics</td><td>Assignment</td><td colspan="2">Case study on Real time applications</td><td colspan="3">10 Sessions</td></tr>
<tr><td colspan="8">Introduction to Big Data: Basics of Distributed File System, Four Vs, Drivers for Big data, Big data applications, Structured, unstructured, semi-structured and quasi structured data. Big data Challenges-Traditional versus big data approach.
The Hadoop: History of Hadoop-Hadoop use cases, The Design of HDFS, Blocks and replication management, Rack awareness, HDFS architecture, HDFS Federation, Name node and data node, Anatomy of File write, Anatomy of File read. Role of Data Scientist - Role of Data Analyst – Data Analytics in Product development - Business Intelligence vs Data analytics - Real time Business Analytical ProcessCase studies related to big data applications</td></tr>
<tr><td>Module 2</td><td>Hadoop MapReduce Framework</td><td>Assignment</td><td colspan="2">Installation of multimode cluster</td><td colspan="3">10 Sessions</td></tr>
<tr><td colspan="8">MapReduce : Overview and Need of Distributed processing for big data- Introduction to hadoop framework and MapReduce programming - HDFS design and its goals - Master-Slave Architecture of hadoop – Working with hadoop daemons-Installation of hadoop single node cluster and multi node clusters - Working with MapReduce programming.</td></tr>
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Module 3	Hive and Hbase Analytical tools	Term paper/Assignment	Hive joins	10 Sessions

Hive : Apache Hive with Hive Installation, Hive Data Types, Hive Table partitioning, Hive DDL commands, Hive DML commands, and Hive sort by vs. order by, Hive Joining tables, Hive bucketing.

Hbase : Introduction to HBase and its working architecture- Commands for creation and listing of tables-disabled and is disabled of table - enable and is enabled of table- describing and dropping of table-Put and Get command - delete and delete all command-commands for scan, count, truncate of tables.

Module 4	Data Analytics with Spark	Term paper/Assignment	Spark RDD	10 Sessions

Spark: Spark: Apache Spark's Philosophy, History of Spark, Running Spark, A Gentle Introduction to Spark, Spark's Basic Architecture, Spark Applications, DataFrames, Partitions, Transformations, Lazy Evaluation, Actions, Spark UI, An End-to-End Example, Integration of Hive and spark.

Nosql: Mongo DB: Introduction ,Features ,Data types , Mongo DB Query language , CRUD operations ,Arrays , Functions: Count ,Sort , Limit , Skip , Aggregate , Cursors – Indexes , Mongo Import , Mongo Export.

List of Laboratory Tasks

Introduction to Hadoop Ecosystem tools

Introduction to Hadoop distributed file System.

Installation of Hadoop single node cluster using Ubuntu operating system.

Working with Hadoop Commands

Introduction to Mapreduce framework

Word Count analysis using sample data set (MapReduce)

Stock analysis using sample data set (MapReduce)

Web log analysis using sample data set (MapReduce)

Temperature analysis using sample data set .(MapReduce)

Working on basic hive commands

Working on basic hbase commands

Install, Deploy & configure Apache Spark

Word count analysis using RDD and FlatMap

Working with MongoDB using restaurant data.

Targeted Application & Tools that can be used:

Apache Hadoop-

HDFS – for data storage

Map reduce – Mapping and reducing.

Hive – Structured data,HQl

Hbase, MongoDB – No SQL

Apache Spark – SCALA LANGUAGE